

HAMZA MERHABENE

GIS Analyst | Surveying Engineer | Geomatics & Renewable Energy Geospatial Support
R Bernardo Lima 47, 1esq, 1150-075 Lisboa, Portugal | +351 932238935 | merhbenhamza07@gmail.com |
[linkedin.com/in/merhabene-hamza-b388a6219](https://www.linkedin.com/in/merhabene-hamza-b388a6219)

PROFESSIONAL SUMMARY

Surveying Engineer and GIS Analyst with 4+ years of experience in GIS analysis, spatial analysis, site suitability investigations, constraints mapping, topographic surveying, drone photogrammetry, LiDAR processing, and remote sensing. Proficient in ArcGIS Pro, QGIS, and Python/ArcPy, with a track record of producing accurate, compliant map outputs and maintaining GIS databases for infrastructure and land development projects. Experienced in cadastral and land subdivision work, geospatial data collection and integration, and cross-functional collaboration with multidisciplinary teams. Strong interest in supporting wind, solar, and energy storage development through reliable, well-documented geospatial data and analysis.

CORE SKILLS

GIS & Spatial Analysis: ArcGIS Pro, QGIS, FME, Global Mapper, site suitability analysis, constraints mapping, thematic mapping, topological error correction, coordinate transformation, datum conversion

Land & Cadastral: Land subdivision, cadastral surveying, land tenure and title cross-referencing, GIS database development and maintenance, data integrity and accessibility

Surveying: GNSS/GPS RTK, Total Station, Geomax, leveling, staking, earthwork and stock volume calculation

CAD & Drafting: AutoCAD, AutoCAD Map 3D, Covadis, MicroStation, GIS-to-CAD conversion, cartographic styling

Photogrammetry & LiDAR: Pix4D, Agisoft Metashape, Erdas, TerraScan, GCP identification, georeferencing, point cloud classification, DSM/DTM generation, orthomosaic production

Remote Sensing: ENVI, satellite imagery analysis, land cover classification

Databases & Programming: PostgreSQL, PostGIS, SQL, Python, ArcPy, HTML5, CSS3, JavaScript

Web GIS: Lizmap, QField

Languages: Arabic (Native), French (Fluent), English (Fluent)

PROFESSIONAL EXPERIENCE

GIS Consultant (Freelance) – Land & Tenure Data Analysis

Oct 2025 – Present

Independent Contractor | Vancouver, British Columbia, Canada (Remote)

- Produce thematic maps, cartographic layouts, and technical documentation supporting site suitability, land analysis, and tenure assessment across 15+ project areas.
- Conduct constraints mapping by cross-referencing geological, cadastral, and tenure datasets (PLSS, BLM, Crown concessions, LTSA, mining claims), reducing due-diligence review time by approximately 40%.
- Convert GIS data to CAD and produce professional AutoCAD plans with complete cartographic styling, managing coordinate transformations and datum conversions in ArcGIS Pro.
- Automate repetitive geoprocessing and file management tasks using Python and ArcPy; build custom ArcGIS toolboxes that cut manual processing time by an estimated 60%.

Cartographic and Photogrammetric Studies Officer

Oct 2023 – Oct 2025

WOLO | Centre Urbain Nord, Tunis, Tunisia

- Performed GCP identification and georeferencing for drone imagery across 30+ survey missions supporting infrastructure and land development projects.
- Generated dense point clouds, digital surface models (DSM), and digital terrain models (DTM) using ArcGIS Pro, covering an estimated 500+ hectares of terrain.
- Produced orthomosaics and 3D mesh models and conducted spatial analysis and thematic mapping to support territorial planning and site assessment decisions.
- Processed and classified LiDAR point clouds (ground, vegetation, buildings) using MicroStation and TerraScan; generated DTM/DSM outputs and performed noise filtering.
- Ensured quality control of cartographic deliverables, maintaining accuracy and compliance standards across all client submissions.

Surveying and GIS Engineering Intern (Final Year Project)

Mar 2023 – Sep 2023

GeoTop | Nabeul, Tunisia

- Designed a low-voltage electrical network GIS database for STEG using UML modeling and PostgreSQL, ensuring data integrity and accessibility.
- Collected field data for all elements of a low-voltage electrical network using QField, capturing 1,000+ network assets.
- Processed survey data and corrected topological errors using ArcGIS and ArcPy; integrated collected data into the production database using SQL.
- Implemented a Web GIS application with Lizmap for network visualization and asset management.

Topographic Surveying Intern

Jun 2021 – Aug 2021

Topography BUT, But Groupe | Tunis, Tunisia

- Conducted topographic surveys and calculated stock volumes in preparation for quarry exploitation.
- Performed land subdivision and plotting of lots for land development projects.
- Processed survey data and produced topographic maps using AutoCAD and Covadis; calculated earthwork volumes for roads and platforms according to project design lines.

ACADEMIC PROJECTS

- **Flood Risk Mapping, ESAT University (Oct 2022 – Jan 2023):** Performed spatial analysis of flood-prone areas using remote sensing and GIS; produced risk maps supporting territorial planning strategies.
- **Real-Time Parcel Tracking System, ESAT University (Feb 2022 – May 2022):** Studied and designed a delivery tracking application integrating real-time geolocation features.

EDUCATION

Master's Degree in Data Science

Universidade Lusófona, Lisboa, Portugal | 2026 – 2027 | ulusofona.pt/en/lisboa/masters/data-science

Master's Degree (Diplome National d'Ingenieur) in Geomatics and Surveying

ESAT University, Tunis, Tunisia | Sep 2020 – Oct 2023 | Graduated with Honors

Bachelor of Technology in Civil Engineering, Specialization in Surveying and Geospatial Information Systems

ISSET de Nabeul, Tunisia | Sep 2017 – Jun 2020 | Graduated with Honors

Tunisian Baccalaureate in Technology

Lycee Mohamed Boudhina, Hammamet, Tunisia | Sep 2016 – Jul 2017

CERTIFICATIONS

- Geospatial Programming: Python (ArcPy – ArcGIS) — Udemy